Quaternion Algebras

Preliminary Arizona Winter School, 2026

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Introduction

Welcome!

These notes about quaternion algebras are written for the Preliminary Arizona Winter School in 2026 and aimed at an audience of undergraduates with a first course in abstract algebra. If that does not describe you, then that doesn't mean these notes aren't for you! However, you may want to brush up on a few concepts from linear and abstract algebra including bases, dimension, ideals, ring homomorphisms, field extensions, and finite fields. Algebra texts such as [DF04, Chapters 7,8] or [Sil22, Chapters 3–5] are a great place to start.

The notes are meant to be self contained, but I hope that they will spark your curiosity and you will feel inspired to explore this beautiful subject more deeply. Some textbook references that you might find helpful as a starting point include [GS17, Ser73, Voi21], although all of these are at the graduate level. Another source freely available online is [Conb] which was also invaluable to me while writing these notes.

Scope

Quaternion algebras are a vast subject. These notes take a somewhat narrow perspective with an eye towards the topic of the 2026 Arizona Winter School: Rationality and Irrationality: Arithmetic and Geometry. These notes introduce quaternion algebras from an algebraic perspective with a goal of distinguishing between quaternion algebras over the rational numbers via arithmetic and geometry. Five lectures is short! There are many aspects of quaternion algebras that we must skip. One that may be of interest to the casual reader is their connections to visualizations; see for example [Han06, Chapters 1 and 2].

What will we cover?

Here is a brief roadmap of what we'll cover in the five lectures:

- **Lecture 1 (Hamilton quaternions and foundations):** We start with Hamilton's quaternions $\mathbb{H}$ as historical motivation before generalizing to arbitrary quaternion algebras $(a, b)_F$ over a base field F.
- **Lecture 2 (Isomorphisms, bases, and splitting):** We characterize when two quaternion algebras are isomorphic and explore the concept of *splitting*—determining when $(a, b)_F \cong M_2(F)$ versus when it remains a division algebra.
- **Lecture 3 (Central simple algebras & associated conics):** We place quaternion algebras into the broader context of central simple algebras (CSAs) and establish a connection between algebra and geometry by associating a conic curve to each quaternion algebra.
- **Lecture 4 (A primer on p-adics):** We introduce the p-adic integers $\mathbb{Z}_p$, p-adic valuations to give us the machinery needed to understand behavior of quaternion algebras over the rational numbers in the final lecture.
- **Lecture 5 (Quaternion algebras over $\mathbb{Q}$):** We define a tool called the Hilbert symbol $(a, b)_p$ and apply a local-global principle to distinguish non-isomorphic quaternion algebras over $\mathbb{Q}$.

How should you use these notes?

As many others have probably said before me, mathematics is not a spectator sport! Even reading mathematics should be active rather than passive. When you encounter a new definition, try working through an example. Challenge yourself to come up with your own example or instructive non-example. Additionally, you should make some time to work through the inline exercises scattered throughout the text (which are distinct from the problem sets accompanying the course).

What if you find a typo?

I hope there won't be many, but inevitably there will be some. Just send me an email. I appreciate it!

Acknowledgements

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Lecture 1: Hamilton quaternions and foundations

1.1 Motivation

The construction of the complex numbers $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}\}$ provides a bridge between algebra and geometry. In the 2-dimensional plane, addition of complex numbers corresponds to vector addition or translation, while multiplication encodes scaling and rotations. For instance, multiplying a vector by i corresponds to a $90°$ rotation.

Given the correspondence between planar rotations and multiplication in the complex numbers, a natural question arises. Can we construct a similar 3-dimensional algebra to model rotations in 3-dimensional space? William Rowan Hamilton spent over a decade searching for such a system (from roughly 1830 until his breakthrough on October 16, 1843). He eventually realized that a 3-dimensional associative division algebra over $\mathbb{R}$ is an algebraic impossibility. To preserve the ability to divide by non-zero elements while extending the complex numbers, we must make a structural compromise. By expanding the construction to a 4-dimensional algebra and dropping the axiom of commutativity, Hamilton arrived at the quaternions, denoted $\mathbb{H}$. This algebra serves as our archetypal example of a non-commutative division ring and forms the foundation for the general theory of quaternion algebras we will develop in this course.

1.2 Hamilton's quaternions

Definition 1.1: Hamilton Quaternions

The **(Hamilton) quaternions** *are*

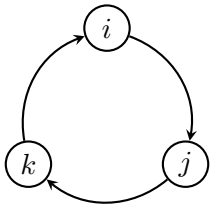
$$\mathbb{H} \coloneqq \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\},$$

with multiplication defined by the relations

$$i^2 = j^2 = k^2 = -1, \quad ij = k = -ji, \quad \text{and every } a \in \mathbb{R} \text{ commutes with } i, j, k.$$

Just as the complex numbers form a 2-dimensional $\mathbb{R}$-vector space with basis $\{1, i\}$, the Hamilton quaternions form a 4-dimensional $\mathbb{R}$-vector space spanned by $\{1, i, j, k\}$. Equipped with the multiplication rules above, $\mathbb{H}$ becomes an associative ring. A helpful way to remember the rules

for multiplication in $\mathbb{H}$ is to arrange i, j, k clockwise in a circle. Moving along the arrows gives a positive product (e.g., $ij = k$, $jk = i$, $ki = j$), while moving against the arrows gives a negative product (e.g., $ji = -k$, $kj = -i$, $ik = -j$).



The rules for multiplication among i, j, k together with the distributive law are all we need to multiply any elements in $\mathbb{H}$.

Example 1.2. $(i + 2j)(2i - j) = 2i^2 - ij + 4ji - 2j^2 = -2 - k - 4k - (-2) = -5k$.

Before proving structural properties, it is helpful to get some hands-on practice calculating directly with elements in $\mathbb{H}$.

Exercise 1.3. Let $q = 1 + 2i - j$ and $r = 2i + k$ be quaternions in $\mathbb{H}$.

(i) Compute the products qr and rq. Verify explicitly that $qr \neq rq$.

(ii) Calculate q^2 and r^2.

(iii) Compute $(q + r)(q - r)$ and compare it to $q^2 - r^2$. Why are these two expressions not equal in $\mathbb{H}$?

To get a better sense of how non-commutativity controls arithmetic in $\mathbb{H}$, we can ask: which quaternions still commute with every element, and how many square roots of -1 exist?

Exercise 1.4.

(i) The **center** of $\mathbb{H}$ is defined as $Z(\mathbb{H}) = \{q \in \mathbb{H} \mid qx = xq \text{ for all } x \in \mathbb{H}\}$. Prove that $Z(\mathbb{H}) = \mathbb{R}$. *(Hint: Write $q = a + bi + cj + dk$ and compute $qi - iq$ and $qj - jq$.)*

(ii) Find all **pure quaternions**, i.e., elements of the form $q = bi + cj + dk$, that satisfy $q^2 = -1$. How does your result compare with the complex numbers $\mathbb{C}$?

The Hamilton quaternions $\mathbb{H}$ are built over the real numbers with $i^2 = j^2 = -1$. However, we can abstract this construction by replacing $\mathbb{R}$ with an arbitrary base field F, and allowing i^2 and j^2 to equal any non-zero scalars $a, b \in F^\times$. While $\mathbb{H}$ is optimized for geometry over $\mathbb{R}$, extending

this framework to other fields including the rational numbers or finite fields turns out to give us a powerful tool in number theory.

In these settings, quaternion algebras provide arithmetic machinery used to study quadratic forms, prime factorization, and the classification of non-commutative rings. This leads us directly to our general definition.

1.3 General quaternion algebras

From now on, F is assumed to be field with characteristic not equal to 2.

Definition 1.5: Quaternion algebra

Let F be a field and let $a, b \in F^\times$. The **quaternion algebra** $(a,b)_F$ *is the 4-dimensional F-algebra with basis $\{1, i, j, ij\}$ and multiplication determined by the relations*

$$i^2 = a, \quad j^2 = b, \quad ij = -ji.$$

Remark 1.6. *Let's clarify two pieces of terminology we'll use throughout these notes:*

- *An F-***algebra** *A is both an F-vector space and a ring, where scalar multiplication and ring multiplication are compatible, i.e.,*

$$(cx)(dy) = (cd)(xy) \quad \text{for all } c, d \in F \text{ and } x, y \in A.$$

- *A* **division algebra over** *F is an F-algebra D in which every non-zero element $v \in D$ has a multiplicative inverse $v^{-1} \in D$.*

Note: A division algebra is simply a field if multiplication is commutative, or a non-commutative "skew field" like $\mathbb{H}$ if it is not.

Aside 1.7. *Why do we enforce* $\mathrm{char}(F) \neq 2$*? Recall that in a field of characteristic 2, we have $1 + 1 = 0$, which implies $1 = -1$. If we allowed* $\mathrm{char}(F) = 2$*, our core non-commutative relation $ij = -ji$ would become $ij = ji$. The algebra would become a commutative ring! It is worth noting that quaternion algebras* do *exist in characteristic 2, but they require a completely different set of structural relations (namely, $i^2 + i = a$, $j^2 = b$, and $ji = (i+1)j$) to preserve non-commutativity. We will leave the characteristic 2 setting aside for this course, but the interested reader should consult [GS17, Section 1.2] for further details.*

Example 1.8 (Quaternion algebras over $\mathbb{Q}$)**.** *By varying the parameters $a, b \in \mathbb{Q}^\times$, we obtain different 4-dimensional algebras over the rational numbers:*

- *In $(-1,-1)_{\mathbb{Q}}$, we have $i^2 = -1$ and $j^2 = -1$. This is simply Hamilton's quaternions restricted to rational coordinates $a, b, c, d \in \mathbb{Q}$.*
- *In $(2,3)_{\mathbb{Q}}$, we have $i^2 = 2$, $j^2 = 3$, and $ij = -ji$. An element looks like $v = v_0 + v_1 i + v_2 j + v_3 ij$ with $v_k \in \mathbb{Q}$, where $ij \cdot ij = (ij)(-ji) = -i(jj)i = -i(3)i = -3i^2 = -6$.*

To study the structure of general quaternion algebras, we need an algebraic mechanism to extract scalar quantities from quaternion elements. Recall that for a complex number $z = x + iy$, multiplying by its complex conjugate $\overline{z} = x - iy$ yields a real number, $|z|^2 = x^2 + y^2$, which measures its squared magnitude. We can generalize this construction to any quaternion algebra $(a,b)_F$ by defining a conjugation map that allows us to recover an element of the base field F via multiplication, giving us an algebraic analogue of squared magnitude known as the *norm*.

Definition 1.9: Conjugate and Norm

Let $v = v_0 + v_1 i + v_2 j + v_3 ij$ be an element of the quaternion algebra $(a,b)_F$. The **conjugate** *of v is*

$$\overline{v} \coloneqq v_0 - v_1 i - v_2 j - v_3 ij$$

and the **norm** *of v is*

$$N(v) \coloneqq v\overline{v} = v_0^2 - a v_1^2 - b v_2^2 + ab v_3^2.$$

The map $(a,b)_F \to (a,b)_F$ defined by $v \mapsto \overline{v}$ is called an **anti-automorphism** of the F-algebra $(a,b)_F$, which means it is an F-vector space isomorphism of $(a,b)_F$ satisfying $vw \mapsto \overline{vw} = \overline{w}\,\overline{v}$. Moreover $\overline{\overline{v}} = v$, and an anti-automorphism with this property is called an **involution** in ring theory. For more information about involutions and central simple algebras, see [KMRT98].

Proposition 1.10: Invertible elements in quaternion algebras

An element q of the quaternion algebra $(a,b)_F$ is invertible if and only if it has nonzero norm. In particular, $(a,b)_F$ is a division algebra if and only if the norm $N\colon (a,b)_F \to F$ does not vanish except at 0.

Proof. The norm is a multiplicative function since

$$N(q_1 q_2) = q_1 q_2 \overline{(q_1 q_2)} = q_1 q_2\ \overline{q_2}\ \overline{q_1} = q_1 N(q_2) \overline{q_1} = q_1 \overline{q_1} N(q_2) = N(q_1) N(q_2).$$

(Since $N(q_2) \in F$ is a scalar in the base field, it commutes with the quaternion $\overline{q_1}$.)

If $qq' = 1$ for some element q' then $N(q)N(q') = N(1) = 1$ in F thus $N(q) \neq 0$. Conversely, if $N(q) \neq 0$, the quaternion $\overline{q}/N(q)$ is a two-sided multiplicative inverse for q. □

Proposition 1.10 provides a straightforward algebraic criterion for invertibilty. To see this in action, we can ask whether the norm can actually vanish for a non-zero element, or if every quaternion algebra is automatically a division algebra. The following example demonstrates that the norm can indeed vanish, even over familiar fields like the rational numbers.

Example 1.11. *Consider the quaternion algebra $A = (1, -1)_{\mathbb{Q}}$, where $i^2 = 1$ and $j^2 = -1$. Let's examine the element $v = 1 + i$. Its norm is*

$$N(1 + i) = (1 + i)(1 - i) = 1 - i + i - i^2 = 1 - 1 = 0.$$

Since $N(v) = 0$ but $v \neq 0$, Proposition 1.10 tells us that $1 + i$ is not invertible. In fact, it is a zero divisor! Thus, $(1, -1)_{\mathbb{Q}}$ is an example of a quaternion algebra that is not a division algebra.

Exercise 1.12. For each of the following quaternion algebras, determine whether it is a division algebra or not. Justify your answer either by demonstrating that the norm map has no non-trivial zeros or by explicitly constructing a non-zero zero divisor.

(i) $A_1 = (-1, -1)_{\mathbb{R}}$ (The classical Hamilton quaternions $\mathbb{H}$)

(ii) $A_2 = (2, 2)_{\mathbb{Q}}$

(iii) $A_3 = (1, 5)_{\mathbb{Q}}$

(iv) Come up with your own examples!

Hint: For (i), inspect the signs of the coefficients in the norm form. For (ii) and (iii), write out the norm form and look for small integer choices for v_0, v_1, v_2, v_3 that cause the terms to cancel out to zero.

Looking Ahead

In this introductory lecture, we have established the foundational structure of quaternion algebras and observed that their algebraic structure is closely tied to the behavior of the quadratic norm form $N(v)$. Specifically, Proposition 1.10 shows that whether an algebra admits a division ring structure depends entirely on whether the equation $N(v) = 0$ has non-trivial solutions.

In the next lecture, we will show that the presence of these non-zero zero divisors implies that the algebra is isomorphic to the matrix ring $M_2(F)$, a property known as being *split*. This sets up the central theme of the course: classifying quaternion algebras up to isomorphism over various base fields. We will achieve this by translating our algebraic questions into a geometric setting, where finding zero divisors corresponds to finding solutions to an equation that defines a curve. From there, we will explore the behavior of quaternion algebras over finite fields, and eventually introduce specialized number systems called p-adic fields that allow us to study arithmetic one prime number at a time. Ultimately, we will see how combining this "localized" information allows us to determine the behavior of a quaternion algebra over the rational numbers.

Key Takeaways

Non-commutativity: Extending the complex numbers to a 3-dimensional associative division algebra over $\mathbb{R}$ is not possible. To build a higher-dimensional analogue of $\mathbb{C}$ with these properties, we must expand to 4 dimensions and drop commutativity ($ij = -ji$).

Algebra is governed by arithmetic: A general quaternion algebra $(a, b)_F$ is a division ring if and only if its associated norm form $N(v) = v_0^2 - av_1^2 - bv_2^2 + abv_3^2$ has no non-trivial zeros.

Lecture 2: Isomorphisms, bases, and splitting

To begin classifying quaternion algebras, we must first make precise what it means for two algebras to have the same structure. Because these algebras are built directly over a base field F, a meaningful structural mapping must preserve not only the ring operations but also the underlying scalar action of F. This motivates our formal definition of an algebra isomorphism.

2.1 Isomorphisms of quaternion algebras

Definition 2.1: Isomorphism of quaternion algebras

A **isomorphism** *between two (quaternion) algebras A and A' over F is a ring isomorphism $\varphi\colon A \xrightarrow{\sim} A'$ that fixes the elements of F, i.e., $\varphi(c) = c$ for all $c \in F$.*

It turns out that the isomorphism class of the quaternion algebra $(a, b)_F$ depends only on the classes of a and b in $F^\times/F^{\times 2}$. In order to make sense of this statement, we can use a particular type of basis to show that two quaternion algebras are isomorphic.

Definition 2.2: Quaternionic basis

A basis of $(a, b)_F$ of the form $\{1, u, v, uv\}$ with $u^2 \in F^\times$, $v^2 \in F^\times$, and $uv = -vu$ is called a **quaternionic basis** *of $(a, b)_F$.*

The notion of quaternionic basis gives some immediate isomorphisms between different quaternion algebras. Since the sets below are formed by scaling and swapping the original basis vectors, they automatically remain bases, though you should check for yourself to see how the new parameters emerge!

1. Since $\{1, i, j, ij\}$ is a quaternionic basis for $(a, b)_F$ (where $i^2 = a$, $j^2 = b$ and $ij = -ji$), so too is $\{1, j, i, ji\}$. Thus $(a, b)_F \cong (b, a)_F$.
2. Similarly, $\{1, i, ij, aj\} = \{1, i, ij, i^2 j\}$ is a quaternionic basis for $(a, b)_F$. Hence $(a, b)_F \cong (a, -ab)_F$.
3. Similarly, $\{1, j, ij, -bi\} = \{1, j, ij, jij\}$ is a quaternionic basis for $(a, b)_F$. Hence $(a, b)_F \cong (b, -ab)_F$.

4. Finally, $\{1, ci, dj, cdij\}$ is a quaternionic basis of $(a,b)_F$ for all $c, d \in F^\times$. Thus $(a,b)_F \cong (ac^2, bd^2)_F$ for all nonzero $c, d \in F$.

In particular, note that taking $b = 1$ in (2) above gives $(a,1)_F \cong (a,-a)_F$, while $b = -1$ yields $(a,-1)_F \cong (a,a)_F$. Up to isomorphism, the algebra $(a,b)_F$ depends on a, b only up to multiplication by nonzero squares in F; we have $(a,c^2)_F = (a,1)_F = (1,a)_F$ and $(c^2,b)_F = (1,b)_F$.

We can summarize the discussion above in the following useful corollary.

Corollary 2.3

Let $a, b \in F^\times$. The isomorphism class of $(a,b)_F$ satisfies:

1. $(a,b)_F \cong (b,a)_F$.
2. $(a,b)_F \cong (a,-ab)_F \cong (b,-ab)_F$.
3. $(a,b)_F \cong (ac^2, bd^2)_F$ *for all* $c, d \in F^\times$.
4. $(ca,cb)_F \cong (-ab,ca)_F$ *for all* $c \in F^\times$.

The preceding examples show how changing a known basis alters the parameters a and b. The following exercise demonstrates that, conversely, any four elements satisfying the relations in Definition 2.2 automatically yield a valid basis.

Exercise 2.4. ([Conb]) Given $u, v \in (a,b)_F$ such that u, v are not in F but $u^2 \in F^\times$, $v^2 \in F^\times$, and $uv = -vu$, show that $\{1, u, v, uv\}$ is a linearly independent set and hence is a quaternionic basis of $(a,b)_F$.

(Hint: Recall $\mathrm{char}(F) \neq 2$. One approach is to show first that $\{1, u\}$ is independent. Then $\{1, u, v\}$ using $uv = -vu$ and $u^2 \in F^\times$, and finally $\{1, u, v, uv\}$ using $u^2, v^2 \in F^\times$ and $uv = -vu$.)

The next result shows that $M_2(F)$, the algebra of 2×2 matrices over F, is a quaternion algebra over F. In particular, any quaternion algebra $(a,b)_F$ where a or b is a square (such as $(a,c^2)_F$) is isomorphic to $M_2(F)$.

Proposition 2.5: $M_2(F)$ is a quaternion algebra

For all $a \in F^\times$, there is an isomorphism of F-algbras $(1,a)_F \cong M_2(F)$.

Proof. Let $\{1, i, j, ij\}$ be a basis for $(1, a)_F$. Consider the map $\varphi\colon (1, a)_F \to M_2(F)$ defined by

$$1 \mapsto \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad i \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad j \mapsto \begin{pmatrix} 0 & a \\ 1 & 0 \end{pmatrix}, \quad ij \mapsto \begin{pmatrix} 0 & a \\ -1 & 0 \end{pmatrix},$$

and extend F-linearly to obtain:

$$v_0 + v_1 i + v_2 j + v_3 ij \mapsto \begin{pmatrix} v_0 + v_1 & a(v_2 + v_3) \\ v_2 - v_3 & v_0 - v_1 \end{pmatrix}.$$

The image of $1, i, j, ij$ in $M_2(F)$ forms a linearly independent set of size $4 = \dim(M_2(F))$ so the map is a bijection. It fixes $c \in F$ since $c \mapsto \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix}$. Finally, one can verify that the map φ is multiplicative. □

Exercise 2.6. Verify that the map φ defined in Proposition 2.5 is indeed multiplicative and the image of $1, i, j, ij$ forms a linearly independent set.

Because $M_2(F)$ contains matrices that square to zero (nilpotents) or square to themselves (idempotents), we can use the isomorphism φ to construct explicit zero-divisors and elements satisfying $e^2 = e$ directly inside the quaternion algebra $(1, a)_F$.

Exercise 2.7. Let $\varphi\colon (1, a)_F \xrightarrow{\sim} M_2(F)$ be the isomorphism defined in Proposition 2.5.

(i) Find the unique element $v \in (1, a)_F$ such that $\varphi(v) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$.

(ii) Verify directly in $(1, a)_F$ that $v^2 = 0$.

(iii) Find the unique element $e \in (1, a)_F$ such that $\varphi(e) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, and verify directly that $e^2 = e$.

2.2 Splitting of quaternion algebras

This relationship between a quaternion algebra and a matrix ring is a fundamental structural property that we will study across different fields. We define this behavior explicitly below.

Definition 2.8: Split quaternion algebra

A quaternion algebra over F is called **split** *if it is isomorphic to $M_2(F)$ as an F-algebra.*

Example 2.9. *Up to isomorphism, there are only two quaternion algebras over $F = \mathbb{R}$ and one quaternion algebra over $F = \mathbb{C}$. In particular, since $\mathbb{R}^\times/\mathbb{R}^{\times 2} \cong \{\pm 1\}$,*

$$(a,b)_{\mathbb{R}} \cong \begin{cases} \mathbb{H} = (-1,-1)_{\mathbb{R}}, & \text{if } a \text{ and } b \text{ are negative} \\ M_2(\mathbb{R}), & \text{if } a \text{ or } b \text{ is positive,} \end{cases}$$

and all elements of $\mathbb{C}^\times$ are squares in $\mathbb{C}$, so $(a,b)_{\mathbb{C}} \cong M_2(\mathbb{C})$ for all $a, b \in \mathbb{C}^\times$. That is, all quaternion algebras over $\mathbb{C}$ are split.

Over $F = \mathbb{Q}$, however, the situation is quite different. We shall see that there are infinitely many non-isomorphic quaternion algebras over $\mathbb{Q}$.

So far, we've seen that a simple condition that guarantees $(a,b)_F$ is split is to have $b \in F^{\times 2}$. However, this condition can be weakened to obtain other quaternion algebras that are split. First, we'll need to understand a particular subgroup of elements of $F^\times$.

Lemma 2.10

For $a \in F^\times$, the set of nonzero elements $x^2 - ay^2$ with $x, y \in F$ is a subgroup of $F^\times$.

Recall that if a is not a square in $F^\times$, then $F(\sqrt{a}) = \{x + \sqrt{a}y \mid x, y \in F\}$ is a field and is isomorphic to the quotient $F(\sqrt{a}) \cong F[t]/(t^2 - a)$. The map $N_{F(\sqrt{a})/F} \colon F(\sqrt{a})^\times \to F^\times$ defined by $x + \sqrt{a}y \mapsto x^2 - ay^2$ is a homomorphism of multiplicative groups. Therefore the image $N_{F(\sqrt{a})/F}(F(\sqrt{a})^\times)$ is a subgroup of $F^\times$. We will verify Lemma 2.10 directly, however.

Proof of Lemma 2.10. Let $N_a \coloneqq \{x^2 - ay^2 | x, y \in F\} \setminus \{0\}$. Then $1 \in N_a$ since we can take $x = 1$ and $y = 0$. N_a is closed under multiplication since

$$(x_1^2 - ay_1^2)(x_2^2 - ay_2^2) = (x_1x_2 + ay_1y_2)^2 - a(x_1y_2 + x_2y_1)^2.$$

Finally, N_a is closed under inversion since

$$\frac{1}{x^2 - ay^2} = \frac{x^2 - ay^2}{(x^2 - ay^2)^2} = \left(\frac{x}{x^2 - ay^2}\right)^2 - a\left(\frac{y}{x^2 - ay^2}\right)^2.$$

□

Observe that N_a always contains the subgroup $F^{\times 2}$ of all squares in $F^\times$ by taking $y = 0$. When a is a square, the set N_a contains all of $F^\times$.

Lemma 2.11

If a is a square in F, then $N_a = F^\times$.

Proof. Suppose $a = c^2$ for $c \in F^\times$. Then for any $x, y \in F$ we have

$$x^2 - ay^2 = x^2 - c^2y^2 = (x - cy)(x + cy).$$

Letting $u = x + cy$ and $v = x - cy$ we have $x = \frac{u+v}{2}$ and $y = \frac{u-v}{2c}$ (recall, $\mathrm{char}(F) \neq 2$ and $c \neq 0$ so this is an invertible linear change of variables). For any $b \in F^\times$, set $u = 1$ and $v = b$, gives $x = \frac{1+b}{2}$ and $y = \frac{1-b}{2c}$ so that $x^2 - ay^2 = b$. □

With these lemmas in hand, we are ready to give several equivalent characterizations of splitness.

Theorem 2.12: Characterizations of splitness

The following are equivalent for a quaternion algebra $(a, b)_F$.

1. *The algebra $(a, b)_F$ is split.*
2. *The algebra $(a, b)_F$ is not a division algebra.*
3. *The norm map $N\colon (a, b)_F \to F$ has a nontrivial zero.*
4. *The element b is in the subgroup N_a.*

Proof. The implication (1) $\implies$ (2) is immediate from the definition of split, and (2) $\iff$ (3) is precisely the content of Proposition 1.10.

To prove (3) $\implies$ (4), we may assume that a is not a square in F. So, suppose that $v = v_0 + v_1 i + v_2 j + v_3 ij$ is a nonzero quaternion with $0 = N(v) = v_0^2 - av_1^2 - bv_2^2 + abv_3^2$. That is, $(v_2^2 - av_3^2)b = v_0^2 - av_1^2$. Note that since a is not a square in F, we must have that $v_2^2 - av_3^2$ is nonzero. Thus

$$b = (v_0^2 - av_1^2)(v_2^2 - av_3^2)^{-1} \in N_a,$$

by Lemma 2.10.

Finally, we prove (4) $\implies$ (1). Again, we assume a is not a square in F. If $b \in N_a$ then $b^{-1} \in N_a$, hence there exist elements $r, s \in F$ such that $b^{-1} = r^2 - as^2$. Let $u \coloneqq rj + sij$ and $v \coloneqq (1 + a)i + (1 - a)ui$, we claim that $\{1, u, v, uv\}$ is a quaternionic basis of $(a, b)_F$.

We have

$$u^2 = r^2j^2 + s^2(ij)^2 - rsij^2 + rsij^2 = br^2 - abs^2 = b(r^2 - as^2) = bb^{-1} = 1.$$

Similarly, one can check that $v^2 = 4a^2$, and $uv = -vu$. Note that by Exercise 2.4, linear independence follows. However, we give a sketch of the details here using matrices of linear transformations. Observe that

$$\begin{aligned} 1 &= 1 + 0i + 0j + 0ij \\ u &= 0 + 0i + rj + sij \\ v &= 0 + (1+a)i - as(1-a)j - r(1-a)ij \\ uv &= 0 + (1-a)i - as(1+a)j - r(1+a)ij \end{aligned},$$

Thus the matrix of the transformation is

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1+a & 1-a \\ 0 & r & -as(1-a) & -as(1+a) \\ 0 & s & -r(1-a) & -r(1+a) \end{pmatrix},$$

and one computes that $\det(M) = 4ab^{-1} \neq 0$, hence M is invertible and $\{1, u, v, uv\}$ is a basis for $(a, b)_F$.

Thus the algebra $(a, b)_F$ is isomorphic to the algebra $(1, 4a^2)_F$ which is split by Proposition 2.5, as desired. □

Corollary 2.13

A quaternion algebra $(a, b)_F$ is either a division algebra over F or is isomorphic to $M_2(F)$.

Exercise 2.14. Let $a \in F^\times$ and consider the algebra $(a, -a)_F$ with standard quaternionic basis. Show that $(i + j)^2 = 0$. Use Theorem 2.12 to conclude that $(a, -a)_F \cong M_2(F)$.

Remark 2.15. *Since we have shown that $(a, b)_F \cong (b, a)_F$, all statements in Theorem 2.12 must be symmetric with respect to the parameters a and b. In particular, $b \in N_a$ if and only if $a \in N_b$. That is, the existence of a nontrivial solution to $b = x^2 - ay^2$ in F is equivalent to the existence of a nontrivial solution to $a = x^2 - by^2$ in F! In Lecture 3, we will introduce an alternative characterization of splitness from a geometric perspective, which makes this symmetry more readily apparent. Finally, if a is not a square in F, then $b \in N_a$ is equivalent to the statement that b is a norm from the extension $F(\sqrt{a})/F$.*

Key Takeaways

Square class dependence: The isomorphism class of $(a, b)_F$ depends only on the square classes of its parameters in $F^\times/F^{\times 2}$. Swapping or scaling basis elements reveals immediate structural equivalences, such as $(a, b)_F \cong (ac^2, bd^2)_F$.

A dichotomy: Every quaternion algebra over F is either a division ring or *split*, i.e., isomorphic to $M_2(F)$. The presence of even a single nonzero zero divisor forces the entire algebra to be isomorphic to a matrix ring over F!

Lecture 3: Central simple algebras & associated conics

With some basic properties of quaternion algebras established, we now explore two approaches for studying their deeper structure. The first is algebraic, placing quaternion algebras into the broader landscape of ring theory as the lowest-dimensional examples of *central simple algebras*. This perspective allows us to see how the structural properties of quaternions generalize to a much wider class of non-commutative rings. The second is geometric, translating the algebraic question of whether a quaternion algebra splits into a concrete problem of finding rational points on an associated quadratic curve known as a conic. While these two perspectives are distinct, together they provide a cohesive framework for understanding how quaternion algebras behave.

3.1 Central Simple Algebras

Definition 3.1: Central simple algebra

A finite-dimensional algebra A over a field F is called a **central simple algebra** *if the center of A is exactly F and A contains no nontrivial proper two-sided ideals.*

The theory of central simple algebras is rich and much of it is beyond the scope of this course. The interested reader should consult, for example, [GS17]. However, using the results of this section, we can prove that every quaternion algebra over a field F is a central simple algebra (CSA).

Proposition 3.2: Quaternion algebras are CSAs

Every quaternion algebra over F is a central simple algebra.

Proof. Let $A = (a, b)_F$ be a quaternion algebra over F with standard quaternionic basis $\{1, i, j, ij\}$. We first show that A is central, i.e., $Z(A) = F$. Let $v = v_0 + v_1 i + v_2 j + v_3 ij \in A$. If $v \in Z(A)$, then v commutes with j, i.e.,

$$0 = vj - jv = 2(v_1 ij + v_3 bi).$$

Since $\operatorname{char} F \neq 2$ and the basis elements i and ij are linearly independent over F, we find $v_1 = v_3 = 0$. Similarly, v must also commute with i, so that $0 = iv - vi = 2v_2(ij)$ hence $v_2 = 0$ and $v = v_0 \in F$. Thus A is central.

By Corollary 2.13, A is either a division algebra or isomorphic to a matrix algebra. If A is a division

algebra, then by definition, every nonzero element is invertible. If $I \subseteq A$ is a nonzero two-sided ideal, it contains some nonzero element x. Since x is invertible, $x^{-1} \in A$, hence $1 = x^{-1}x \in I$. Thus, $I = A$, so A has no nontrivial proper two-sided ideals.

If $A \cong M_2(F)$, let $I \subseteq M_2(F)$ be a nonzero two-sided ideal. Then I contains a matrix $M = (m_{ij})$ with at least one nonzero entry. By multiplying on the left or right by the permutation matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, we can swap rows or columns to bring a nonzero entry to the top-left position. Thus, without loss of generality, we assume the m_{11} entry is nonzero. Since I is a two-sided ideal, we can isolate this entry by multiplying by matrices on the right and left:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} m_{11} & 0 \\ 0 & 0 \end{pmatrix} \in I.$$

Since $m_{11} \in F^\times$, scaling by its inverse (i.e., multiplying by the scalar matrix corresponding to m_{11}^{-1}) yields $e_{11} = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right) \in I$. Similarly, one can show that I contains $e_{22} = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. Thus I contains their sum $I_2 = e_{11} + e_{22}$, i.e., the identity matrix, and hence $I = M_2(F)$.

In either case, A contains no nontrivial proper two-sided ideals. Hence, A is simple. □

Since quaternion algebras are CSAs, they fit into a much broader class of non-commutative rings. The Artin-Wedderburn Theorem (cf. [GS17, Theorem 2.1.3] states that any CSA over a field F is isomorphic to a matrix ring $M_n(D)$ over some division algebra D. For a 4-dimensional quaternion algebra A, dimension considerations limit us to only two possibilities. If $n = 1$, then A is isomorphic to a 4-dimensional division algebra D. If $n = 2$, then $M_2(D)$ has dimension 4 and forces $D = F$, meaning $A \cong M_2(F)$ and the algebra splits. Remarkably, this abstract classification recovers a result we have already seen through direct computation in Corollary 2.13: a quaternion algebra must be either a division algebra or isomorphic to $M_2(F)$.

However, we don't need the full machinery of this higher-dimensional classification to analyze the structure of a specific quaternion algebra $(a, b)_F$. While we can determine whether $(a, b)_F$ is split or a division algebra by directly searching for zero divisors via the norm form, this algebraic approach can be computationally intensive. Instead, we can translate the condition for splitting into the language of geometry. By constructing an associated quadratic curve called a conic, we reduce the question of whether the algebra splits to the classical problem of determining if this conic possesses an F-rational point.

3.2 The associated conic

To see why a conic curve emerges in this setting, let's look back at how we detect zero divisors. Recall that by Proposition 1.10 a quaternion algebra $A = (a, b)_F$ splits if and only if it contains a nonzero element with a norm of zero. For a general quaternion $v = v_0 + v_1 i + v_2 j + v_3 ij$, this means finding a nontrivial solution to the four-variable equation:

$$N(v) = v_0^2 - av_1^2 - bv_2^2 + abv_3^2 = 0.$$

Instead of dealing with all four variables simultaneously, we can restrict our attention to the 3-dimensional subspace of **pure quaternions**, defined by:

$$A^0 \coloneqq \{xi + yj + zij \in A \mid x, y, z \in F\}.$$

These are simply the quaternions with scalar term v_0 equal to zero. Consider a general non-zero pure quaternion $q = xi + yj + zij \in A^0$. Its norm simplifies to:

$$N(xi + yj + zij) = -ax^2 - by^2 + abz^2.$$

In order for q to be a zero divisor, its norm must vanish:

$$-ax^2 - by^2 + abz^2 = 0 \implies ax^2 + by^2 = abz^2.$$

Dividing both sides by the non-zero scalar ab yields:

$$\frac{x^2}{b} + \frac{y^2}{a} = z^2 \implies a\left(\frac{y}{a}\right)^2 + b\left(\frac{x}{b}\right)^2 = z^2.$$

Setting $X = \frac{y}{a}$ and $Y = \frac{x}{b}$, this simplifies to $aX^2 + bY^2 = z^2$. For solutions with $z \neq 0$, we can divide by z^2 to obtain an equation of the form $aX^2 + bY^2 = 1$. This motivates the following definition of the associated conic.

Definition 3.3: Associated Conic

Let $a, b \in F^\times$ and let $A \coloneqq (a, b)_F$ be a quaternion algebra. The **associated conic** *$C(a, b)$ is the affine plane curve defined by the equation $ax^2 + by^2 = 1$ We say $C(a, b)$ has an F-rational point if there exists a pair $(x, y) \in F^2$ satisfying this equation.*

Thus, the equation defining the associated conic governs whether a nonzero pure quaternion has

norm zero. Finding an F-rational point on this curve directly constructs a zero divisor, allowing us to translate a ring-theoretic problem into the geometric problem of finding points on a planar curve.

Exercise 3.4. Consider the quaternion algebra $A = (2, -7)_{\mathbb{Q}}$.

(i) Find an explicit $\mathbb{Q}$-rational point (x_0, y_0) on the associated conic $2x^2 - 7y^2 = 1$.

(ii) Use your point (x_0, y_0) to construct an explicit nonzero element $v \in A$ with $N(v) = 0$.

Having seen how a single rational point on $C(a, b)$ constructs an explicit zero divisor, we now formalize this equivalence in full generality. The following theorem establishes that the existence of an F-rational point on the conic is both necessary and sufficient for the algebra to split.

Theorem 3.5: Geometric condition for splitness

For $a, b \in F^\times$ the following conditions are equivalent:

1. $(a, b)_F \cong M_2(F)$
2. *The equation $ax^2 + by^2 = 1$ has a solution (x, y) in F.*
3. *The equation $ax^2 + by^2 = z^2$ has a solution (x, y, z) in F other than $(0, 0, 0)$.*

Proof. We will prove the equivalence of the three conditions by establishing the cycle of implications (2) $\implies$ (3) $\implies$ (1) $\implies$ (2).

(2) $\implies$ (3): If (x, y) is a solution to $ax^2 + by^2 = 1$, then $(x, y, 1)$ is a non-trivial solution to $ax^2 + by^2 = z^2$.

(3) $\implies$ (1): Suppose $ax_0^2 + by_0^2 = z_0^2$ for some $(x_0, y_0, z_0) \neq (0, 0, 0)$. We consider two cases based on y_0. If $y_0 = 0$, then $x_0 \neq 0$ and so $a = \left(\frac{z_0}{x_0}\right)^2$ is a square, so $(a, b)_F$ splits by Theorem 2.12. If $y_0 \neq 0$, rearranging the equation gives $b = \left(\frac{z_0}{y_0}\right)^2 - a\left(\frac{x_0}{y_0}\right)^2$, meaning $b \in N_a$, so $(a, b)_F$ splits by Theorem 2.12.

(1) $\implies$ (2): Suppose $(a, b)_F \cong M_2(F)$, so $(a, b)_F$ is split. By Theorem 2.12, $b \in N_a$, meaning $b = u^2 - av^2$ for some $u, v \in F$. Rearranging this gives $av^2 + b(1)^2 = u^2$. If $u \neq 0$, then $a\left(\frac{v}{u}\right)^2 + b\left(\frac{1}{u}\right)^2 = 1$, so $\left(\frac{v}{u}, \frac{1}{u}\right)$ is a solution to the equation in (2). If $u = 0$, then $b = -av^2$ and one can check that $\left(\frac{1+a}{2a}, \frac{1-a}{2av}\right)$ is a solution. □

With Theorem 3.5 in hand, testing whether a quaternion algebra splits simplifies to checking if a specific conic curve has points over F. Rather than wrestling with matrix representations or 4-variable norm equations, we can often read off splitness directly by inspecting the curve's geometry or identifying explicit F-rational points. The following corollaries illustrate how powerful this geometric condition can be in practice.

Theorem 3.6: Splitting of quaternion algebras over a finite field

Let $\mathbb{F} = \mathbb{F}_q$ be a finite field. Every quaternion algebra $(a, b)_{\mathbb{F}}$ over $\mathbb{F}$ is split.

Proof. By Theorem 3.5, it suffices to show that the associated conic $C(a, b)$ has an $\mathbb{F}$-rational point. Recall that the multiplicative group $\mathbb{F}^\times$ of the finite field $\mathbb{F}$ is cyclic of order $q-1$ (see, e.g., [DF04, Section 9.5, Proposition 18]). Thus there are $(q+1)/2$ squares in $\mathbb{F}$, namely all the even powers of the generator of $\mathbb{F}^\times$ and zero. The sets $A \coloneqq \{ax^2 : x \in \mathbb{F}\}$ and $B \coloneqq \{1 - by^2 : y \in \mathbb{F}\}$ therefore both have cardinality $(q+1)/2$. If the equation $ax^2 + by^2 = 1$ had no solutions, then $A \cap B = \emptyset$. But this is impossible since then $\#(A \cup B) > q = \#\mathbb{F}$. Hence $C(a, b)$ must have an $\mathbb{F}$-rational point and thus $(a, b)_{\mathbb{F}}$ is split. □

If the ideas in this proof are new to you, it's worth working through the argument carefully. The next exercise applies the same principles to congruences modulo an odd prime p. This will be useful to us later as well.

Exercise 3.7. Let p be an odd prime and let $a, b \in (\mathbb{Z}/p\mathbb{Z})^\times$.

(i) Show that the set $S_1 = \{x^2 \pmod p \mid x \in \mathbb{Z}/p\mathbb{Z}\}$ has exactly $\frac{p+1}{2}$ elements.

(ii) Show that the set $S_2 = \{b + ay^2 \pmod p \mid y \in \mathbb{Z}/p\mathbb{Z}\}$ also has exactly $\frac{p+1}{2}$ elements.

(iii) Use the pigeonhole principle to conclude that $S_1 \cap S_2 \neq \emptyset$, and deduce that the congruence $x^2 - ay^2 \equiv b \pmod p$ always has a solution for any choice of units a and b.

Certain algebraic relations between the parameters a and b automatically guarantee that a quaternion algebra splits. We state these special cases below and leave their verification as an exercise.

Corollary 3.8

Let F be a field with $\operatorname{char} F \neq 2$.

1. *For any $a \in F^\times \setminus \{1\}$, the quaternion algebra $(a, 1-a)_F$ splits.*
2. *For any $a \in F^\times$, the quaternion algebra $(a, -a)_F$ splits.*

Exercise 3.9. Prove Corollary 3.8 by applying Theorem 3.5 to find explicit F-rational points on the associated conics $ax^2 + (1-a)y^2 = 1$ and $ax^2 - ay^2 = 1$. Can you find other algebraic approaches to prove these statements?

Corollary 3.10

Let F be a field with $\operatorname{char} F \neq 2$. *The quaternion algebra $\mathbb{H}(F) := (-1,-1)_F$ is a division ring if and only if -1 is not the sum of two squares in F.*

Proof. We prove the contrapositive. By Theorem 3.5, $(-1,-1)_F \cong M_2(F)$ if and only if the equation $-x^2 - y^2 = 1$ has a solution (x, y) in F, i.e., if and only if $x^2 + y^2 = -1$ has a solution in F. Since a quaternion algebra is either split or a division algebra, the contrapositive gives the result. □

We've seen that the relationship between F-rational points on $C(a, b)$ and zero-divisors in $(a, b)_F$ is not just theoretical. It allows us to construct explicit zero-divisors and parameterize rational points. The following exercise explores the latter.

Exercise 3.11. Suppose the associated conic $C(a, b)\colon ax^2 + by^2 = 1$ has a known F-rational point $P = (x_0, y_0)$.

(i) Consider the line $y - y_0 = t(x - x_0)$ through P with slope $t \in F$. Substitute this into $ax^2 + by^2 = 1$ to find the second intersection point $(x(t), y(t))$ as rational functions of t.

(ii) Conclude that if F is an infinite field and $(a, b)_F$ splits, the conic $C(a, b)$ contains infinitely many F-rational points.

Key Takeaways

Quaternion algebras are CSAs: Every quaternion algebra over a field F is a central simple algebra (CSA), meaning its center is exactly F and it contains no nontrivial proper two-sided ideals. This places quaternions into the broader landscape of ring theory as the lowest-dimensional examples of CSAs.

Geometric criterion for splitness: The algebraic problem of checking whether a quaternion algebra splits reduces completely to a 2-variable geometric problem. The algebra $(a, b)_F$ splits if and only if its associated plane conic $C(a, b) : ax^2 + by^2 = 1$ has an F-rational point

Lecture 4: A primer on p-adics

In the previous lecture, we established a useful geometric characterization of splitting: a quaternion algebra $(a,b)_F$ splits if and only if the associated conic curve $C(a,b)\colon ax^2+by^2=1$ has an F-rational point. When working over a general field F, finding F-rational points on $C(a,b)$ can be quite difficult. Over $\mathbb{Q}$, for instance, determining whether a conic has rational points might require testing infinitely many fractions. However, testing for solutions over other related fields containing $\mathbb{Q}$ like $\mathbb{R}$ or the p-adic numbers $\mathbb{Q}_p$ turns out to be much more straightforward. Over $\mathbb{R}$, for instance, $ax^2+by^2=1$ has a solution if and only if $a>0$ or $b>0$. Over $\mathbb{Q}_p$, the algebraic structure limits the possible square classes, allowing us to determine the existence of solutions using finitely many calculations. This motivates a powerful strategy: rather than approaching problems directly over $\mathbb{Q}$, we can study them "locally" over $\mathbb{R}$ and over each p-adic field $\mathbb{Q}_p$.

As we will see in Lecture 5, a quaternion algebra over $\mathbb{Q}$ splits globally if and only if it splits "locally" everywhere. To make this precise, we must first construct the p-adic numbers $\mathbb{Q}_p$. We give a brief introduction to p-adic numbers in this section from an algebraic (rather than topological) perspective. Entire books have been written on the subject, e.g., [Gou20, Kat07, Pol24], and the interested reader is strongly encouraged to explore beyond these notes. See also Catherine Hsu's PAWS notes from 2023 [Hsu24] and Keith Conrad's notes [Cona].

4.1 The p-adic integers

We begin by formalizing the inverse limit construction, which provides an algebraic framework needed to build the p-adic integers $\mathbb{Z}_p$.

4.1.1 Construction via inverse limits

Definition 4.1: Inverse Limits

An **inverse system** *of rings is a collection of rings $\{A_n\}_{n\geq 1}$ with homomorphisms $\{f_n\}_{n\geq 1}$:*

$$\dots A_{n+1} \xrightarrow{f_n} A_n \xrightarrow{f_{n-1}} A_{n-1} \dots \xrightarrow{f_2} A_2 \xrightarrow{f_1} A_1.$$

The **inverse limit** $A \coloneqq \varprojlim A_n$ *is the subset of $\prod_n A_n$ given by*

$$A = \{(a_n) \in \textstyle\prod_{n\geq 1} A_n \mid f_n(a_{n+1}) = a_n \text{ for all } n \geq 1\}.$$

For each $n \geq 1$, the projection map $\pi_n\colon A \to A_n$ sends $a \mapsto a_n$.

For more details on inverse limits, see for example, [DF04, Section 7.6].

Definition 4.2: p-adic integers

For any prime number p, the ring of **p-adic integers** *is $\mathbb{Z}_p \coloneqq \varprojlim \mathbb{Z}/p^n\mathbb{Z}$ of the inverse system of rings $\{\mathbb{Z}/p^n\mathbb{Z}\}_{n\geq 1}$ and homomorphisms $f_n \colon \mathbb{Z}/p^{n+1}\mathbb{Z} \to \mathbb{Z}/p^n\mathbb{Z}$ defined by reduction modulo p^n.*

Exercise 4.3. Show that f_n is well-defined. That is, given a residue class $\bar{x} \in \mathbb{Z}/p^{n+1}\mathbb{Z}$, show that any choice of representative $x \in \bar{x}$ maps to the same residue class in $\mathbb{Z}/p^n\mathbb{Z}$. Further show that $\mathbb{Z}_p$ is indeed a ring with addition and multiplication defined componentwise; given two elements of $\mathbb{Z}_p$, $x \coloneqq (x_n)_{n\in\mathbb{Z}^+}$ and $y \coloneqq (y_n)_{n\in\mathbb{Z}^+}$, show that $(x+y)_n = x_n + y_n$ and $(x \cdot y)_n = x_n \cdot y_n$ for all $n \geq 1$.

The multiplicative identity in $\mathbb{Z}_p$ is $1 = (\bar{1}, \bar{1}, \bar{1}, \dots)$ where a $\bar{1}$ in the nth index denotes the residue class of 1 in $\mathbb{Z}/p^n\mathbb{Z}$. The map that sends each $x \in \mathbb{Z}$ to $(\bar{x}, \bar{x}, \bar{x}, \dots) \in \mathbb{Z}_p$ is a ring homomorphism with trivial kernel since 0 is the only integer that is congruent to 0 modulo p^n for all $n \geq 1$. Thus $\mathbb{Z}_p$ is a ring with characteristic 0 that contains $\mathbb{Z}$. In order to drop the residue class notation, we may uniquely represent elements of $\mathbb{Z}/p^n\mathbb{Z}$ by $\{0, 1, \dots, p^n - 1\}$. We shall see that $\mathbb{Z}_p$ contains much more than just the ring $\mathbb{Z}$.

Remark 4.4. *While the notation $\varprojlim \mathbb{Z}/p^n\mathbb{Z}$ looks abstract, an element $a \in \mathbb{Z}_p$ is simply an infinite sequence of compatible residue classes $a = (a_1, a_2, a_3, \dots)$ where $a_n \in \mathbb{Z}/p^n\mathbb{Z}$ and $a_{n+1} \equiv a_n \pmod{p^n}$. To make this more concrete, consider a square root of -1 in $\mathbb{Z}_5$, represented by an element $\alpha \in \mathbb{Z}_5$ with $\alpha^2 = -1$. One such root is given by the sequence $(2, 7, 57, 182, \dots)$, where each term squares to -1 in its respective ring:*

$$2^2 \equiv -1 \pmod 5, \qquad 7^2 = 49 \equiv -1 \pmod{25}, \qquad 57^2 = 3249 \equiv -1 \pmod{125}, \quad \dots$$

Importantly, these terms are compatible under projection down the sequence, e.g.:

$$182 \equiv 57 \pmod{125}, \qquad 57 \equiv 7 \pmod{25}, \qquad 7 \equiv 2 \pmod 5.$$

Thus, $\mathbb{Z}_p$ provides a formal setting where we can solve equations by solving them modulo p^n for all $n \geq 1$ simultaneously.

Example 4.5. *In the ring $\mathbb{Z}_5$ we have*

- $2 = (2, 2, 2, 2, 2, \dots)$
- $-1 = (4, 24, 124, 624, \dots)$

- $170 = (0, 20, 45, 170, 170, \dots)$
- $\frac{1}{2} = (3, 13, 63, 313, \dots)$
- *The two solutions to* $x^2 = -1$ *in* $\mathbb{Z}_5$ *are* $x_1 = (2, 7, 57, 182, \dots)$ *and* $x_2 = (3, 18, 68, 443, \dots)$
- $\sqrt[3]{2} = (3, 3, 53, 303, 2178, \dots)$

Why do we care about $\sqrt{-1} \in \mathbb{Z}_5$?

Recall from Lecture 3 that a quaternion algebra $(a, b)_F$ splits if and only if it contains a zero divisor if and only if it contains an element of norm 0. Let $\mathbb{Q}_5$ be the field of fractions of $\mathbb{Z}_5$ (see Exercise 4.9 Definition 4.11). For the algebra $(-1, -1)_{\mathbb{Q}_5}$, the norm of the element $q = xi + yj$ is $N(q) = x^2 + y^2$. Since -1 is a square in $\mathbb{Z}_5$, we can pick $\alpha \in \mathbb{Z}_5$ with $\alpha^2 = -1$. Then we find

$$N(\alpha i + j) = \alpha^2(1) + 1^2 = -1 + 1 = 0.$$

The existence of $\sqrt{-1} \in \mathbb{Z}_5$ directly yields a zero divisor in $(-1, -1)_{\mathbb{Q}_5}$, forcing the algebra to split over $\mathbb{Q}_5$. In general, the ring $\mathbb{Z}_p$ provides a structured setting where solving norm equations reduces to solving congruences modulo powers of p!

4.1.2 Representations and p-adic expansions

Although representing elements of $\mathbb{Z}_p$ as sequences (a_n) with $a_n \in \mathbb{Z}/p^n\mathbb{Z}$ is natural from the definition of $\mathbb{Z}_p$ as an inverse limit, it is also somewhat redundant. In particular, the value of a_n constrains the possible value of a_{n+1} since $a_{n+1} \equiv a_n \bmod p^n$. If we uniquely represent each a_n as an integer in the set $\{0, \dots, p^n - 1\}$, then we can always write $a_{n+1} = a_n + p^n b_n$ with $b_n \in \{0, \dots, p-1\}$.

Definition 4.6: p-adic expansion of a p-adic integer

Let $a = (a_n) \in \mathbb{Z}_p$ *be a* p*-adic integer with each* a_n *uniquely represented by an integer in* $\{0, \dots, p^n - 1\}$. *Then the sequence* $(b_0, b_1, b_2, \dots)$ *with* $b_0 = a_1$ *and* $b_n = (a_{n+1} - a_n)/p^n$ *is called the* p-**adic expansion** *of* a.

It follows from the definition that every element of $\mathbb{Z}_p$ has a unique p-adic expansion and every sequence of integers $(b_0, b_1, b_2, \dots)$ of integers in $\{0, \dots, p-1\}$ is the p-adic expansion of some element of $\mathbb{Z}_p$. Indeed, we can recover (a_n) from its p-adic expansion via $a_1 = b_0$ and $a_{n+1} = a_n + p^n b_n$ for all $n \geq 1$.

Thus, there is a bijection between $\mathbb{Z}_p$ and the set of all sequences of integers in $\{0, \dots, p-1\}$

indexed by the non-negative integers. We can represent elements of $\mathbb{Z}_p$ as formal power series in powers of p, i.e., $(b_0, b_1, b_2, \dots)$ can be expressed as $\sum_{n\geq 0} b_n p^n$.

Remark 4.7. *Note the distinction between the two representations of an element in $\mathbb{Z}_p$:*

- *As an inverse limit, $a = (a_1, a_2, a_3, \dots)$ consists of compatible residue classes $a_n \in \mathbb{Z}/p^n\mathbb{Z}$. For example, in $\mathbb{Z}_5$ we have $\frac{1}{2} = (3, 13, 63, 313, \dots)$*
- *As a p-adic expansion, $a = \sum_{n\geq 0} b_n p^n$ records the individual p-adic digits $b_n \in \{0, \dots, p-1\}$. For example, in $\mathbb{Z}_5$ we have $\frac{1}{2} = 3 + 2 \cdot 5 + 2 \cdot 5^2 + 2 \cdot 5^3 + \dots$*

These are related by $a_n \equiv \sum_{i=0}^{n-1} b_i p^i \pmod{p^n}$.

Example 4.8. *Let's revisit Example 4.5. We have the following 5-adic expansions in $\mathbb{Z}_5$:*

- $2 = (2, 0, 0, 0, 0, \dots) = 2 + 0 \cdot 5 + 0 \cdot 5^2 + 0 \cdot 5^3 + \dots$
- $-1 = (4, 4, 4, 4, \dots) = 4 + 4 \cdot 5 + 4 \cdot 5^2 + 4 \cdot 5^3 + \cdots = \sum_{n\geq 0} 4 \cdot 5^n$
- $170 = (0, 4, 1, 1, 0, 0, \dots) = 0 + 4 \cdot 5 + 1 \cdot 5^2 + 1 \cdot 5^3$
- $\frac{1}{2} = (3, 2, 2, 2, \dots) = 3 + 2 \cdot 5 + 2 \cdot 5^2 + 2 \cdot 5^3 + \cdots = 3 + \sum_{n\geq 1} 2 \cdot 5^n$
- *The two solutions to $x^2 = -1$ are $x_1 = 2 + 1 \cdot 5 + 2 \cdot 5^2 + 1 \cdot 5^3 + 3 \cdot 5^4 + \dots$ and $x_2 = 3 + 3 \cdot 5 + 2 \cdot 5^2 + 3 \cdot 5^3 + 1 \cdot 5^4 + \dots$*
- $\sqrt[3]{2} = 3 + 0 \cdot 5 + 2 \cdot 5^2 + 2 \cdot 5^3 + 3 \cdot 5^4 + \dots$

The expansion of -1 might seem surprising at first glance, so here's another way to think about it. Note that -1 is by definition the additive inverse of 1. The additive inverse $x \coloneqq \sum_{n\geq 0} b_n 5^n$ of 1 must satisfy the property that $1 + x = 0$, i.e., $1 + b_0 + b_1 5 + b_2 5^2 + \cdots = 0$. Thus $b_0 + 1 = 0$ and since $b_0 \in \{0, \dots, 4\}$, we must have $b_0 = 4$. So $5 + b_1 5 + b_2 5^2 + \cdots = 0$. But then $b_1 + 1 = 0$, so $b_1 = 4$. Continuing in this fashion we see that $b_n = 4$ for all $n \geq 0$.

4.1.3 Divisibility and the fraction field $\mathbb{Q}_p$

You might wonder about what other properties the ring $\mathbb{Z}_p$ possesses. One important fact is that the ring contains no zero divisors.

Exercise 4.9. Prove that $\mathbb{Z}_p$ is an integral domain.

Unfortunately, however, $\mathbb{Z}_p$ is not a field. Not every p-adic integer has a multiplicative inverse, but we can characterize those elements that do.

Exercise 4.10. Prove that a p-adic integer $x = \sum_{n\geq 0} b_n p^n$ is invertible if and only if $b_0 \neq 0$. In other words, $x \in \mathbb{Z}_p$ is invertible if and only if x is not divisible by p.

Since $\mathbb{Z}_p$ is an integral domain but not a field, it's natural to study its field of fractions.

Definition 4.11: p-adic field

The **field of p-adic numbers** $\mathbb{Q}_p$ *is the field of fractions of* $\mathbb{Z}_p$.

As a fraction field, the elements of $\mathbb{Q}_p$ are by definition all pairs $(a, b) \in \mathbb{Z}_p^2$ typically written as a/b, modulo the equivalence relation $a/b \sim c/d$ whenever $ad = bc$. However, it turns out that we can represent elements of $\mathbb{Q}_p$ more explicitly by extending our notion of a p-adic expansion to allow negative indices, with the condition that only finitely many p-adic digits with negative indices are nonzero. If we view p-adic expansions in $\mathbb{Z}_p$ as formal power series in p, then in $\mathbb{Q}_p$ elements are represented as formal Laurent series in p. We'll give a proof of this fact shortly.

Exercise 4.12.

(i) Construct a solution $(x, y) \in \mathbb{Q}_5^2$ with $x, y \neq 0$ to $x^2 + y^2 = -1$, and conclude (again) that $(-1, -1)_{\mathbb{Q}_5} \cong M_2(\mathbb{Q}_5)$.

(ii) Show that $x^2 \equiv -1 \pmod 3$ has no solution in $\mathbb{Z}/3\mathbb{Z}$, and deduce that -1 is not a square in $\mathbb{Q}_3$. Use the fact that there exists $\beta \in \mathbb{Z}_3$ satisfy $\beta^2 = -2$ (see Lemma 5.7 in the next lecture for a way to show this element exists) to find an explicit solution $(x, y) \in \mathbb{Q}_3^2$ and conclude that $(-1, -1)_{\mathbb{Q}_3} \cong M_2(\mathbb{Q}_3)$.

(iii) Explain in one sentence why $x^2 + y^2 = -1$ has no solutions in $\mathbb{R}$, illustrating why $(-1, -1)_{\mathbb{R}}$ remains a non-split division algebra.

4.2 p-adic valuations

In order to better understand the field $\mathbb{Q}_p$, we need to introduce a new notion of size on the set of rational numbers. Let $p \in \mathbb{Z}^+$ be a prime number.

Definition 4.13: p-adic valuation

The p-**adic valuation** *on* $\mathbb{Q}$ *is the function* $v_p\colon \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ *defined as follows. For nonzero* $x \in \mathbb{Z}$*, there is a unique integer* $v_p(x)$ *with the property that* $x = p^{v_p(x)}u$ *with* $p \nmid u$*. We define* $v_p(0) = \infty$ *and extend* v_p *to* $\mathbb{Q}$ *by defining* $v_p(a/b) = v_p(a) - v_p(b)$*. The* p-**adic norm** *of* $x \in \mathbb{Q}$ *is*

$$|x|_p := \begin{cases} p^{-v_p(x)}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0. \end{cases}$$

Example 4.14. $\left|\dfrac{3 \cdot 2^5}{1+2^5}\right|_2 = 2^{-5}, \qquad |7^{-9}|_2 = 1, \qquad \left|\dfrac{2}{21}\right|_7 = 7^{-(-1)} = 7, \qquad |25!|_5 = 5^{-6}$

Exercise 4.15. Prove that for $a, b \in \mathbb{Z}^+$, $a \equiv b \bmod p^n \iff |a-b|_p \leq p^{-n}$.

Theorem 4.16: Properties of the p-adic valuation

The p*-adic valuation* v_p *satisfies the following properties.*

1. $v_p(x) = \infty$ *if and only if* $x = 0$.
2. $v_p(xy) = v_p(x) + v_p(y)$.
3. $v_p(x+y) \geq \min\{v_p(x), v_p(y)\}$.

Exercise 4.17. Prove Theorem 4.16.

We can extend the definition of the p-adic valuation (and the p-adic norm) to the ring $\mathbb{Z}_p$ and hence to the field $\mathbb{Q}_p$. Let $x \in \mathbb{Z}_p$. Suppose that the smallest power of p dividing x is $k \geq 0$:

$$x = \sum_{i \geq k} b_i p^i = b_k p^k + b_{k+1}p^{k+1} + \cdots = p^k(b_k + b_{k+1}p + \dots) = p^k \sum_{n \geq 0} b_{n+k} p^n.$$

Then $v_p(x) = k$. If $k = 0$, i.e., $p \nmid x$, then $v_p(x) = 0$. That is, the p-adic valuation on $\mathbb{Z}_p$ gives the smallest power of p in the p-adic expansion of an element.

Note that the p-adic valuation on $\mathbb{Z}_p$ and Theorem 4.16 gives us another way to quickly verify that $\mathbb{Z}_p$ is an integral domain. If $x, y \in \mathbb{Z}_p$ are both nonzero then $v_p(x), v_p(y) \neq \infty$, hence $v_p(xy) = v_p(x) + v_p(y) < \infty$ and thus $xy \neq 0$.

Exercise 4.18. The group of p-adic units $\mathbb{Z}_p^\times$ is the multiplicative group of invertible elements

in $\mathbb{Z}_p$. Show that $\mathbb{Z}_p^\times = \{a \in \mathbb{Z}_p \mid v_p(a) = 0\}$. Then show that every nonzero $a \in \mathbb{Z}_p$ can be uniquely written in the form $p^n u$ with $n \geq 0$ and $u \in \mathbb{Z}_p^\times$.

Theorem 4.19: p-adic expansion of elements in $\mathbb{Q}_p$

Every element $x \in \mathbb{Q}_p$ can be represented as a formal Laurent series in p,

$$x := \sum_{n \geq k} b_n p^n = b_k p^k + b_{k+1} p^{k+1} + \cdots + b_0 + b_1 p + b_2 p^2 + \ldots,$$

with $k \in \mathbb{Z}$ and $0 \leq b_n \leq p-1$ for all n.

Proof. If $x \in \mathbb{Z}_p$ then we're done. So assume $x = \frac{a}{b}$ with $a, b \in \mathbb{Z}_p$ and $b \neq 0$. Then $m := v_p(x) = v_p(a) - v_p(b)$, with $v_p(a), v_p(b) \geq 0$ since $a, b \in \mathbb{Z}_p$. Note that m could be negative. Let $u = p^{-m}x$. Then $v_p(u) = v_p(p^{-m}) + v_p(x) = -m + m = 0$. By Exercise 4.18, $u \in \mathbb{Z}_p^\times \subset \mathbb{Z}_p$ and therefore u has a p-adic expansion $u = \sum_{n \geq 0} u_n p^n$. Since $x = p^m u$ we have

$$x = p^m \sum_{n \geq 0} u_n p^n = \sum_{n \geq 0} u_n p^{n+m} = \sum_{n \geq m} u_{n-m} p^n,$$

which is an expression of x in the desired form. □

Aside 4.20. *You might wonder which elements of $\mathbb{Z}_p$ have an eventually periodic expansion. It turns out this precisely characterizes $x \in \mathbb{Q}$. See Keith Conrad's notes [Cona].*

Key Takeaways

Structure of $\mathbb{Z}_p$: The ring of p-adic integers $\mathbb{Z}_p = \varprojlim \mathbb{Z}/p^n\mathbb{Z}$ consists of compatible sequences of residue classes modulo powers of p. Solving equations in $\mathbb{Z}_p$ corresponds to solving them modulo p^n for all $n \geq 1$ simultaneously. An element $x \in \mathbb{Z}_p$ is a unit ($x \in \mathbb{Z}_p^\times$) if and only if $p \nmid x$.

p-adic valuations: The p-adic valuation $v_p \colon \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ measures size by tracking divisibility by p. Extending v_p to $\mathbb{Z}_p$ recovers the group of units as $\mathbb{Z}_p^\times = \{u \in \mathbb{Z}_p \mid v_p(u) = 0\}$, allowing every element of the fraction field $\mathbb{Q}_p$ to be uniquely expressed as $p^m u$ for some $u \in \mathbb{Z}_p^\times$ and $m \in \mathbb{Z}$.

Splitting: Deciding whether a quaternion algebra splits over a field reduces to finding rational points on an associated conic. Since p-adic point-finding boils down to checking solvability modulo powers of p, verifying local splitting becomes a straightforward calculation compared to working over $\mathbb{Q}$, as shown by using $\sqrt{-1} \in \mathbb{Z}_5$ to show that $(-1,-1)_{\mathbb{Q}_5}$ splits.

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Bibliography

[Cona] Keith Conrad. The p-adic expansion of rational numbers. Expository notes. URL: https://kconrad.math.uconn.edu/blurbs/gradnumthy/rationalsinQp.pdf.

[Conb] Keith Conrad. Quaternion algebras. Expository notes. URL: https://kconrad.math.uconn.edu/blurbs/ringtheory/quaternionalg.pdf.

[DF04] David S. Dummit and Richard M. Foote. *Abstract Algebra*. John Wiley & Sons, Inc., Hoboken, NJ, 3rd edition, 2004.

[Gou20] Fernando Q. Gouvêa. *p-adic Numbers: An Introduction*. Universitext. Springer, 3rd edition, 2020.

[GS17] Philippe Gille and Tamás Szamuely. *Central Simple Algebras and Galois Cohomology*, volume 165 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2nd edition, 2017. doi:10.1017/9781316661277.

[Han06] Andrew J. Hanson. *Visualizing Quaternions*. The Morgan Kaufmann Series in Interactive 3D Technology. Morgan Kaufmann, 2006.

[Hsu24] Catherine Hsu. Local fields. Lecture notes for the Preliminary Arizona Winter School (PAWS), 2024. URL: https://swc-math.github.io/aws/2025/PAWSHsu/2024PAWSHsuNotes.pdf.

[Kat07] Svetlana Katok. *p-adic Analysis Compared with Real*, volume 37 of *Student Mathematical Library*. American Mathematical Society, Providence, RI, 2007.

[KMRT98] Max-Albert Knus, Alexander Merkurjev, Markus Rost, and Jean-Pierre Tignol. *The Book of Involutions*, volume 44 of *American Mathematical Society Colloquium Publications*. American Mathematical Society, Providence, RI, 1998. With a preface by J. Tits.

[Pol24] Paul Pollack. *Unreal Analysis: Visions of the p-adic Realm*. Ross Mathematics Program, 2024.

[Ser73] Jean-Pierre Serre. *A Course in Arithmetic*, volume 7 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1973.

[Sil22] Joseph H. Silverman. *Abstract Algebra: An Integrated Approach*, volume 55 of *Pure and Applied Undergraduate Texts*. American Mathematical Society, Providence, Rhode Island, 2022.

[Voi21] John Voight. *Quaternion Algebras*, volume 288 of *Graduate Texts in Mathematics*. Springer, Cham, 2021. Available open access. doi:10.1007/978-3-030-56694-4.